Xujiang Tang

Mathematics, Mathematical Modeling, and Theoretical Machine Learning
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RESEARCH PROFILE

Mathematics and Applied Mathematics undergraduate and Research Assistant at HKUST(GZ). Research spans pure and applied mathematics, mathematical modeling, dynamical systems, optimization, and theoretical machine learning, with particular interests in online learning, bandit algorithms, reinforcement learning, and sequential decision-making.

EDUCATION

B.Sc. in Mathematics and Applied Mathematics — Sep 2023 - Jun 2027
Yangtze University College of Arts and Sciences, Jingzhou, China
- Core strengths: Mathematical Modeling 96; Ordinary Differential Equations 89; Complex Analysis 90; Real Analysis 95; Functional Analysis 99; Abstract Algebra 98.
- Additional preparation in probability and mathematical statistics, optimization, numerical computation, and programming.

2026 AI4Math Summer School: Lean Formalized Mathematics — Jul 2026
Zhejiang University, Hangzhou, China
- Intensive training in Lean-based formal proof, theorem formalization, and computer-assisted mathematical reasoning.

RESEARCH APPOINTMENT AND EXPERIENCE

Research Assistant — 2026 - 2027
The Hong Kong University of Science and Technology (Guangzhou) | Prof. Tianyuan Jin
- Research on online learning, multi-armed bandits, reinforcement learning, and sample- and resource-efficient sequential decision-making.
- Developing mathematical foundations in concentration, martingales, KL divergence, exponential families, and information-theoretic lower bounds.

Algebraic Topology and Polyhedral Products — 2026
Ongoing research discussions and guidance from Prof. Stephen D. Theriault, University of Southampton
- Studying Whitehead products, moment-angle complexes, graph Lie algebras, quasi-Lie structures, and low-prime homotopy theory.
- Developing background in prime-local methods, low-characteristic operations, completion methods, and PBW-type arguments.

Bilevel Optimization and Implicit Differentiation — 2025 - Present
Independent Research
- Investigated hypergradient computation beyond invertible-Hessian assumptions using differentiable lower-level solution manifolds.
- Developed a curvature-filter framework with broad applicability across AI model classes; submitted to *JMLR* (Theory and Methods).

ADDITIONAL RESEARCH EXPERIENCE

Deep Hedging under Rough Volatility — 2025 - Present
Independent Research
- Developed a fractional-kernel neural architecture for variance-optimal hedging under rough-volatility models.
- Integrated stochastic analysis, neural approximation, and numerical optimization.

Dynamical Systems and Koopman Operator Learning — Mar 2024 - Present
Research Collaborator, Prof. Yanbing Jia's Group
- Proposed a Koopman Operator Linearization Skeleton for long-horizon nonlinear prediction.
- Implemented and evaluated the framework on multiple chaotic-system benchmarks against deep-learning baselines.

Peking University Open-Source LLM Group — Jun - Aug 2024
Research and Engineering Intern
- Contributed to large-language-model deployment, retrieval-augmented generation, and external knowledge integration.

PUBLISHED AND ACCEPTED

1. Q. Li and **X. Tang**. HKGEduRec: A Knowledge Graph-Enhanced Dynamic Hybrid Framework for Educational Recommendation with Cold-Start Mitigation. *IEEE ICMEIM 2025*, pp. 1-5. EI-indexed.
2. **X. Tang** and Q. Li. Logical Gene Encoding: A Bio-Inspired Approach for Energy-Efficient Automated Reasoning. *IEIT 2025*. EI-indexed. DOI: 10.2991/978-94-6463-803-5_81.
3. Q. Li and **X. Tang**. Robust Optimal Reinsurance and Investment with Inflation Risk: A Game-Theoretic Approach and Explicit Solutions. *AIMS Mathematics*. Accepted.
4. **X. Tang et al.** Structure-preserving Koopman Predictive Control for Memristive Neural Dynamics: Input-exact and Commutator-defect Lifting. *Biological Cybernetics*. Accepted.

UNDER REVIEW AND SUBMITTED

1. **X. Tang et al.** The Curvature Filter in Bilevel Optimization: Implicit Differentiation Without Nondegeneracy. *Journal of Machine Learning Research*, Theory and Methods. Submitted.
2. Y. Fan, **X. Tang**, and R. Guan. Deep Hedging under Rough Volatility: A Fractional Kernel Embedding Approach with Optimal Convergence Rate. *AIMS Mathematics*. Under review.
3. **X. Tang et al.** Spectral Network Determinants of Seizure-like Synchronization and Spread in Coupled Hindmarsh-Rose Brain Models. *Cognitive Neurodynamics*. Under review.

SELECTED TECHNICAL NOTES

Eight online Markdown notes on sequential decisions: regret and exploration; adaptive data and filtrations; concentration bounds; martingales and optional stopping; KL divergence; exponential-family bandit models; and change-of-measure lower bounds. Available at txj2006.github.io/notes/.

HONORS

Kaggle Silver Medal, Jigsaw Toxic Comment Classification Challenge | Regional Gold Medal, WorldQuant International Quant Championship | Third Prize, UCAS Graduate AI Forum | National Second Prize, National College Students Statistical Modeling Competition | National Second Prize, Hua Zhong Cup Mathematical Modeling Competition

SKILLS

Mathematics: mathematical modeling, real and complex analysis, functional analysis, differential equations, abstract algebra, algebraic topology, probability, and optimization.
Programming and tools: Python, PyTorch, NumPy, Pandas, C++, R, MATLAB, Lean, Git/GitHub, Linux, LaTeX.
Machine learning: online learning, bandit algorithms, reinforcement learning, stochastic modeling, LLM deployment, and retrieval-augmented generation.
Languages: Mandarin Chinese; English.